Homework 11: Orthogonal Projections

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Definition. *Projections*. Suppose that V is a vector space and $P: V \to V$ is a linear transformation.

a) If $P^2 = P$ then P is called a ***projection***.

b) If V is an inner product space and $P^2 = P = P^*$ then P is called an ***orthogonal projection***. We furthermore say that P ***projects onto*** range(P).

Here $*$ in P^* denoted the conjugate transpose, also known as Hermitian transpose of $m \times n$ complex matrix P. There are several notations for conjugate transpose, such as P^H or $P^\dagger$ († reads as dagger and is standard notation used in physics). For real matrices P, the conjugate transpose is just the transpose $P^* = P^T$.

Problem. *Determining if a Matrix is a Projection.*

Determine which of the following matrices are projections. If they are projections, determine whether or not they are orthogonal and describe the subspace of $\mathbb{R}^n$ that they project onto.

$$P = \begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix}, \qquad Q = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 1/2 \\ 0 & 0 & 0 \end{bmatrix}, \qquad R = \begin{bmatrix} 5/6 & 1/6 & -1/3 \\ 1/6 & 5/6 & 1/3 \\ -1/3 & 1/3 & 1/3 \end{bmatrix}.$$

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Good Luck